ADULT HYPOTHERMIA ALGORITHM

GUIDELINES 2025
EUROPEAN RESUSCITATION COUNCIL©

Core temperature < 35° C or cold to touch

Vital signs present?

YES

Impaired consciousness?

YES

NO

Transport to nearest hospital if injured; consider onsite or hospital treatment if uninjured

Hypothermia I[c]

- Warm environment and dry clothing
- Warm sweet drinks
- Active movement

- Heart rate < 45 min^{-1}
- SBP < 90 mmHg[b]
- Cardiocirculatory instability
- Core temperature < 32°C in old and multimorbid or < 30°C in young and healthy

NO TO ALL

Transport to nearest appropriate hospital

YES TO ANY

Transport to hospital with ECLS[d]

Hypothermia II or III[c]

- Minimal and cautious movements to avoid rescue collapse
- Prevent further heat loss
- Active external and minimally invasive rewarming techniques[e]
- Airway management as required

NO

Witnessed hypothermic cardiac arrest

- Obvious signs of death[a]
- Valid do not attempt CPR order
- Conditions unsafe for rescuer
- Avalanche burial > 60 min, airway packed with snow and asystole

YES TO ANY

Consider termination of CPR

NO TO ALL

- Start CPR, do not delay transport
- If continuous CPR is not possible, consider intermittent or delayed CPR in difficult or dangerous rescue
- Airway management
- Core temperature < 30°C max 3 defibrillations, consider IV 1mg adrenaline
- Gather information of mechanism of accident

Cardiac arrest from alternative cause prior to cooling

- Avalanche burial < 60 min

YES

Transport to nearest appropriate hospital or manage as per supervising healthcare provider

NO

Start CPR

- **Transport to hospital with ECLS[d]**
- **Do not terminate CPR**

Consider prognostication to determine benefit of ECLS[f]

- HOPE survival probability ≥10

NO

Consider termination of CPR

YES

Hypothermia IV[c]

- Rewarm with ECLS
- If ECLS not available within 6hrs, CPR and non-ECLS rewarming in peripheral hospital
- Rewarm to core temperature ≥ 32°C

NO ROSC

Cardiac instability resolved

- **Prepare for multi-organ failure and need for ECLS respiratory support**
- **Post-resuscitation care**

CPR - cardiopulmonary resuscitation; ECLS - extracorporeal life support; IV - intravenous; ROSC - return of spontaneous circulation; SBP - systolic blood pressure; [a] Decapitation; truncal transection; whole body decomposed or whole body frozen solid (chest wall not compressible); [b] Systolic blood pressure < 90 mmHg is a reasonable prehospital estimate of cardiocirculatory instability, but for in-hospital decisions, the minimum sufficient circulation for a deeply hypothermic patient (e.g., <28°C) has not been defined; [c] Swiss staging of accidental hypothermia; [d] Direct transport to an ECLS centre is recommended in an arrested hypothermic patient. In remote areas, transport decisions should balance the risk of increased transport time with the potential benefit of treatment in an extracorporeal life support centre. Transfer should commence if such a centre can be reached within 6 hours; [e] Warm environment, chemical, electrical, or forced air heating packs or blankets, and warm IV fluids (38-42°C). In case of cardiac instability refractory to medical management, consider rewarming with extracorporeal life support; [f] If the decision is made to stop at an intermediate hospital to measure serum potassium, a hospital en route to an extracorporeal life support centre should be chosen. The HOPE score should not be used in children, consider expert consultation instead.